

MAKERERE UNIVERSITY BUSINESS SCHOOL
Bachelor of Business Computing
BUC 2225: BUSINESS INTELLIGENCE AND DATA WAREHOUSING
COURSE WORK II — TRIAL EXAMINATION PAPER

Topics Covered: Topic 2: Data Modelling & Database Design | Topic 3: ETL | Topic 4: DW Implementation
Time Allowed: 2 Hours
Instructions: Section A: Answer ALL 30 objective questions (1 mark each).
Section B: Answer ANY THREE structured questions (10 marks each).

SECTION A: OBJECTIVE QUESTIONS (30 Marks)

Instructions: Each question carries ONE mark. Circle or shade the letter of the BEST answer.

1. Data modelling is a key step before any database is built. Which of the following best describes what data modelling involves?
 - a. Writing SQL queries to retrieve data from existing tables
 - b. Developing visual representations of data that will be stored in a database
 - c. Configuring servers and network infrastructure for a system
 - d. Creating backup scripts to protect data from loss
2. Data modelling can be carried out at three distinct levels of abstraction. Which level focuses on what data the system contains without specifying how it will be implemented?
 - a. Physical model
 - b. Logical model
 - c. Conceptual model
 - d. Operational model
3. At a certain level of data modelling, attributes, primary keys, and foreign keys are defined without yet choosing a specific DBMS. Which level of data modelling is being described?
 - a. Conceptual model
 - b. Physical model
 - c. Logical model
 - d. Dimensional model
4. The final stage of data modelling involves specifying exact table names, column names, data types, and DBMS-specific optimisations. Which level of data modelling requires these technical specifications?
 - a. Conceptual model
 - b. Logical model
 - c. Dimensional model
 - d. Physical model
5. ERDs are commonly used tools in database design to plan and communicate data structures. Which of the following best describes the purpose of an ERD?
 - a. A tool used to write and execute SQL statements
 - b. A graphical representation used to visualise relationships between entities in a database

- c. A backup mechanism for relational databases
- d. A report generated from a completed database

6. ERDs use standardised symbols to represent different database components. Which ERD component is represented by a rectangle?

- a. Attribute
- b. Relationship
- c. Entity
- d. Foreign key

7. Entities in an ERD are classified based on whether they can exist independently or rely on another entity. Which statement correctly describes a strong entity?

- a. It depends on another entity for its existence and identification
- b. It has its own unique primary key and exists independently
- c. It is derived from a relationship between two other entities
- d. It cannot participate in more than one relationship

8. Attributes in an ERD come in different types depending on how they hold their values. An 'Address' attribute that can be broken down into Street, City, and Postal Code is an example of which attribute type?

- a. Simple attribute
- b. Derived attribute
- c. Compound (composite) attribute
- d. Multi-valued attribute

9. Some attributes can hold more than one value for the same entity occurrence. Which type of attribute best describes 'Academic Qualifications' for a staff member who holds a Diploma, a Bachelor's degree, and a Master's degree?

- a. Simple attribute
- b. Derived attribute
- c. Composite attribute
- d. Multi-valued attribute

10. Keys are special attributes used to uniquely identify records and establish links between tables. What is the correct term for a candidate key that is NOT selected as the primary key?

- a. Foreign key
- b. Composite key
- c. Alternate key
- d. Super key

11. Foreign keys are essential for maintaining referential integrity between related tables. Which of the following best describes the role of a foreign key in a relational database?

- a. It uniquely identifies each row in its own table
- b. It refers to the primary key of another table to create a relationship
- c. It is always the same as the primary key of its table
- d. It is a key that consists of two or more attributes combined

12. Cardinality in an ERD describes the numerical nature of relationships between entities. In a university system where one Professor teaches many Courses but each Course has only one Professor, what type of cardinality exists?

- a. Many-to-many (M:N)
- b. One-to-one (1:1)
- c. One-to-many (1:M)
- d. Zero-to-many (0:M)

13. Crow's foot notation uses specific symbols to represent different cardinality values on relationship lines. What does the symbol of two parallel lines (double tick marks) represent in crow's foot notation?

- a. Zero or many
- b. One or many
- c. One and only one (mandatory one)
- d. Zero or one

14. Relationship types in ERDs vary based on how entities depend on each other for existence. An Order Line Item entity that cannot exist without a parent Order entity illustrates which type of relationship?

- a. Non-identifying relationship
- b. Recursive relationship
- c. Identifying relationship
- d. Optional relationship

15. A recursive relationship is a special type used when an entity relates to itself. Which real-world example best illustrates a recursive relationship?

- a. A Student enrolling in a Course
- b. An Employee who supervises other Employees
- c. A Product belonging to a Category
- d. A Customer placing an Order

16. Dimensional modelling is a specialised technique used in analytical database environments. Which type of system is dimensional modelling most suited for?

- a. Operational transaction processing systems (OLTP)
- b. Data warehouses and analytical databases (OLAP)
- c. Network management and configuration systems
- d. Real-time embedded operating systems

17. In a dimensional model, two types of tables serve distinct and complementary roles. Which table stores measurable business values such as total revenue, quantity sold, and discount amounts?

- a. Dimension table
- b. Staging table
- c. Metadata table
- d. Fact table

18. Dimension tables provide descriptive context to the measures stored in fact tables. Which of the following would typically be found in a dimension table?

- a. Total sales amount and units sold
- b. Product name, category, and brand
- c. Foreign keys linking to other fact tables
- d. Aggregated revenue figures by quarter

19. The star schema is a widely used design in data warehousing known for its simplicity. Which of the following best describes the structure of a star schema?

- a. Multiple fact tables linked together with no dimension tables
- b. A central fact table directly connected to surrounding dimension tables
- c. Fully normalised tables with many sub-tables at every level
- d. A single flat table containing all attributes and measures

20. The snowflake schema extends the star schema by further normalising dimension tables. Which of the following is a key characteristic of the snowflake schema?

- a. All data is stored in one central fact table only
- b. Dimension tables are split into multiple related sub-tables to reduce redundancy
- c. No relationships exist between the fact table and dimensions
- d. It is always faster to query than a star schema

21. ETL is the core data integration process used to populate a data warehouse. What does the acronym ETL stand for?

- a. Evaluate, Test, Load
- b. Extract, Transform, Load
- c. Export, Transfer, Link
- d. Enter, Translate, Launch

22. The ETL process is divided into three distinct phases, each with a specific responsibility. Which phase is responsible for pulling raw data from source systems such as databases, flat files, and spreadsheets?

- a. Load phase
- b. Transform phase
- c. Extract phase
- d. Staging phase

23. Raw data from source systems often contains inconsistencies, errors, and different formats that must be resolved. During which ETL phase does data cleaning, formatting, and standardisation occur?

- a. Extract phase
- b. Load phase
- c. Backup phase
- d. Transform phase

24. A staging area is an intermediate component that sits between source systems and the data warehouse. What is the primary purpose of using a staging area in the ETL process?

- a. To permanently store historical analytical reports for end users
- b. To temporarily hold raw extracted data before transformation and loading
- c. To serve as the final layer from which users run BI reports

d. To replace the need for a data warehouse altogether

25. The Transform phase covers a wide range of data processing operations before loading. Which of the following is a correct example of a transformation operation?

- a. Copying data directly from the source database to the warehouse without changes
- b. Deleting the source database after extraction is complete
- c. Converting date formats from DD/MM/YYYY to YYYY-MM-DD for consistency
- d. Creating user accounts for analysts who will query the warehouse

26. Data warehouses serve a different purpose from operational databases in an organisation. Which of the following best states the primary objective of a data warehouse?

- a. To process daily transactions such as sales and withdrawals in real time
- b. To replace all operational databases across an organisation
- c. To facilitate efficient querying, analysis, and reporting that supports decision-making
- d. To store network security logs and system configuration files

27. A data warehouse consists of several interconnected components that work together. Which component temporarily holds raw data between the source systems and the warehouse before any transformation takes place?

- a. Metadata repository
- b. Data mart
- c. Staging area
- d. OLAP cube

28. Two dominant data warehouse architectures — proposed by Inmon and Kimball — take different approaches to building a warehouse. Which architecture builds a centralised, normalised data warehouse first as the single source of truth, from which data marts are then derived?

- a. Kimball architecture
- b. Flat architecture
- c. Cloud architecture
- d. Inmon architecture

29. The Kimball architecture follows a bottom-up approach to data warehouse development. Which of the following best describes the Kimball approach?

- a. It starts by building a fully normalised central warehouse before any data marts are created
- b. It builds subject-specific data marts first, which are then integrated into an enterprise warehouse
- c. It stores all data in a single flat table for simplicity
- d. It relies exclusively on cloud infrastructure for storage and processing

30. Implementing a data warehouse follows a structured life cycle from planning to maintenance. During which stage of the Data Warehouse Development Life Cycle (DWDLC) are stakeholder workshops held, business requirements documented, and source data assessed for quality?

- a. Deployment stage
 - b. Testing stage
 - c. Requirements Gathering stage
 - d. Maintenance stage
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SECTION B: STRUCTURED QUESTIONS (30 Marks)

Instructions: Answer ANY THREE (3) questions. Each question carries 10 marks. Be precise and use examples where applicable.

Question 1: Data Modelling and Entity Relationship Diagrams

- (a) Define **data modelling** and explain the THREE levels at which it can be carried out, giving a brief example at each level. (3 marks)
 - (b) Identify and explain FOUR components of an ERD, giving one example of each. (2 marks)
 - (c) Describe the THREE types of cardinality (1:1, 1:M, M:N) used in ERDs and provide one example for each from a school management context. (3 marks)
 - (d) A clinic records Patients (Patient_ID, Name, Date_of_Birth), Doctors (Doctor_ID, Name, Specialisation), and Appointments (Appointment_ID, Date, Time, Diagnosis). Each patient can have many appointments and each appointment is attended by exactly one doctor, but a doctor can attend many appointments. Draw or clearly describe a **logical ERD** using crow's foot notation showing all entities, attributes, keys, and cardinalities. (2 marks)
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Question 2: Dimensional Modelling

- (a) Distinguish between **relational modelling** and **dimensional modelling** in terms of purpose, design approach, and the type of environment each is suited for. (2 marks)
 - (b) Distinguish between a **fact table** and a **dimension table** in a dimensional model and give one real-world example of each. (2 marks)
 - (c) Compare and contrast the **star schema** and the **snowflake schema**. Give TWO advantages and ONE disadvantage of each, and state which you would recommend for a system requiring fast query performance. (3 marks)
 - (d) A bus company operates routes across Uganda and wants to analyse ticket sales by route, time period, ticket type, and boarding station. Design a **star schema** for this system including ONE Fact Table with at least FOUR measures and at least FOUR Dimension Tables with relevant attributes. Clearly label all PKs and FKs. (3 marks)
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Question 3: ETL — Extract, Transform, Load

- (a) Define **ETL** and explain the role of each of its three phases — Extract, Transform, and Load — using a retail supermarket as your context. (3 marks)
- (b) Explain the purpose of a **staging area** in the ETL pipeline and describe TWO specific problems that arise when data is loaded directly into the warehouse without one. (2 marks)
- (c) A hospital data quality report identifies three issues: missing date-of-birth values, duplicate patient records with different name spellings, and age stored as a fixed number instead of being calculated from birth date. For each issue, name the type of data quality problem and describe the transformation technique that should be applied. (3 marks)

(d) Identify and explain TWO common challenges that affect ETL systems at large scale, and suggest one best practice to address each challenge. (2 marks)

Question 4: Data Warehouse Implementation

(a) Identify and explain FIVE key components of a data warehouse, stating the specific role each plays. (2.5 marks)

(b) Describe the Data Warehouse Development Life Cycle (DWDLC) by explaining FIVE of its stages and the key activities carried out at each stage. (2.5 marks)

(c) Compare the **Inmon** and **Kimball** data warehouse architectures in terms of design philosophy, data structure, and speed of delivery. Recommend which is most suitable for: (i) a large government ministry needing a single source of truth, and (ii) a fast-growing startup needing quick analytical results. Justify your recommendations. (3 marks)

(d) Identify and explain TWO challenges commonly faced during data warehouse implementation and propose one best practice to address each. (2 marks)

Question 5: Integrated Case Study

Case Study — KwataKwata Supermarket Chain

KwataKwata is a supermarket chain with 25 branches across Uganda and an online delivery service. The company runs separate systems for point-of-sale transactions, inventory, supplier orders, and a customer loyalty programme. Management has approved a data warehouse project to enable analysis of sales by product category, branch, time period, and customer segment. The project team is currently debating which architecture to adopt.

(a) Identify FOUR data sources KwataKwata would extract data from. For each source, name ONE key data element to be extracted and explain its importance to the warehouse. (2 marks)

(b) Describe the ETL process the team should follow to load sales data into the warehouse. Include at least TWO specific transformation operations with examples from the KwataKwata context. (2 marks)

(c) Design a **star schema** for KwataKwata's sales analysis. Include ONE Fact Table with at least FOUR measures and at least FOUR Dimension Tables with relevant attributes. Label all PKs and FKs clearly. (3 marks)

(d) Recommend ONE data warehouse architecture (Inmon or Kimball) for KwataKwata and provide TWO justified reasons for your recommendation. (2 marks)

(e) Identify TWO challenges KwataKwata is likely to face during implementation and propose one best practice for each. (1 mark)

— END OF EXAMINATION PAPER —

Section A: 30 marks | Section B (best 3): 30 marks | Grand Total: 60 marks